



Lakshya JEE 2.0 (2024)

Limit, Continuity and Differentiability

DPP-06

1. Function $f(x) = (x+1)^{\cot x}$ will be continuous at $x = 0$ if the value of $f(0)$ is

(1) 1 (2) 2
(3) e (4) 3

2. The points at which the function $f(x) = \frac{x+1}{x^2+x-12}$ is discontinuous, are:

(1) -3, 4 (2) 3, -4
(3) -1, -3, 4 (4) -1, 3, 4

3. Function $f(x) = \tan x$ is discontinuous at

(1) $x = 0$ (2) $x = \frac{\pi}{2}$
(3) $x = \pi$ (4) $x = -\pi$

4. The set of all points of discontinuity of $f(x) = x - [x]$ is:

(1) Set of all integers
(2) Set of all irrational numbers
(3) Set of all real numbers
(4) Set of fractional values

5. Let $f: \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} (1 + |\sin x|)^{\frac{3a}{|\sin x|}}, & -\frac{\pi}{4} < x < 0 \\ b, & x = 0 \\ e^{\frac{\cot 4x}{\cot 2x}}, & 0 < x < \frac{\pi}{4} \end{cases}$$

If f is continuous at $x = 0$, then the value of

$$6a + b^2 \text{ is}$$

(1) $1 - e$ (2) $e - 1$
(3) $1 + e$ (4) e

6. If $f(x) = \frac{1}{x-1}$ and $g(x) = \frac{1}{x^2-1}$ then find the points of discontinuity of $f \circ g(x)$

(1) $x = \pm 1, \pm \sqrt{2}$ (2) $x = \pm 2, \pm \sqrt{2}$
(3) $x = \pm 2, \pm \sqrt{3}$ (4) $x = \pm 1, \pm \sqrt{3}$

7. Function $f(x) = |x-1| + |x-2| + \cos x$, where $x \in [0, 4]$ is not continuous at number of points

(1) 3 (2) 2
(3) 1 (4) 0

8. $f(x) = \begin{cases} 2x, & x < 0 \\ 2x+1, & x \geq 0 \end{cases}$. Then

(1) $f(x)$ is continuous at $x = 0$
(2) $f(|x|)$ is continuous at $x = 0$
(3) $f(x)$ is discontinuous at $x = 0$
(4) None of the above

9. Let $f(x) = [x] + \sqrt{x - [x]}$, where $[x]$ denotes the greatest integer function. Then,

(1) $f(x)$ is continuous on $\mathbb{R}^+$
(2) $f(x)$ is continuous on $\mathbb{R}$
(3) $f(x)$ is continuous on $\mathbb{R} - \mathbb{Z}$
(4) None of these

10. Let $f(x) = \begin{cases} 5^{1/x}, & x < 0 \\ \lambda[x], & x \geq 0 \end{cases}$ and $\lambda \in \mathbb{R}$, then at $x = 0$

(1) f is discontinuous
(2) f is continuous only, if $\lambda = 0$
(3) f is continuous only, whatever λ may be
(4) None of the above

11. If $f(x) = \begin{cases} x-1, & x \geq 1 \\ 2x^2-2, & x < 1 \end{cases}$,

$$g(x) = \begin{cases} x+1, & x > 0 \\ -x^2+1, & x \leq 0 \end{cases} \text{ and } h(x) = |x|,$$

then $\lim_{x \rightarrow 0} f(g(h(x)))$ is equal to



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (2)
3. (2)
4. (1)
5. (3)
6. (1)

7. (4)
8. (3)
9. (3)
10. (3)
11. 0



PW Web/App - <https://smart.link/7wwosivoicgd4>

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